

Bayesian regression for language scientists

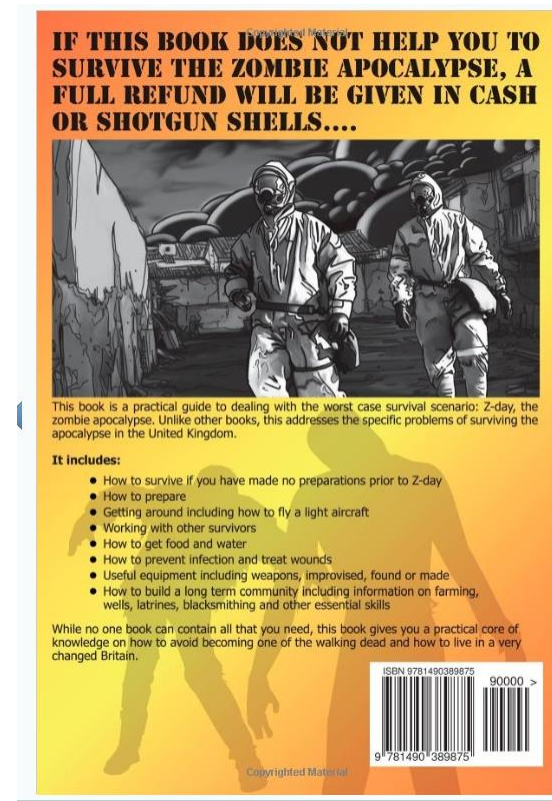
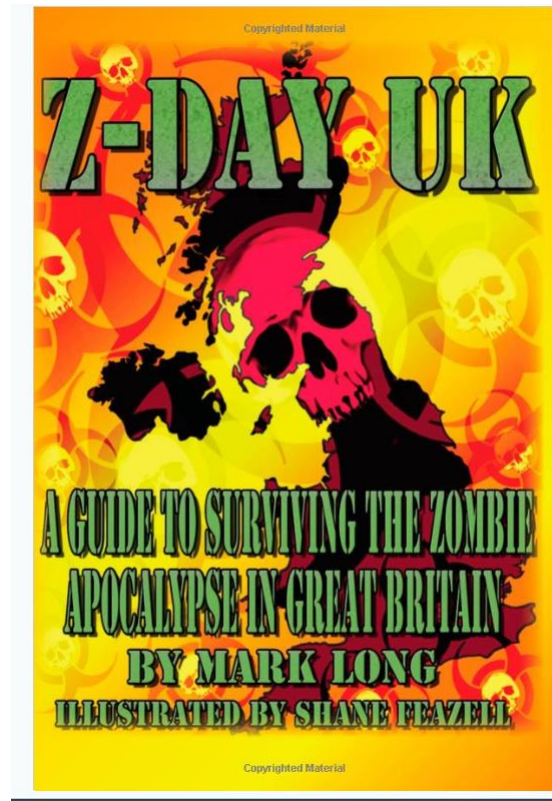
Lecture 3

Natalia Levshina

Radboud University

LOT2026 Summer School

Case study: help + (to) Infinitive



*Here is how items in your emergency kit can **help** you **survive** the zombie apocalypse.*

<https://www.redcross.ca/blog/2013/2/preparedness-101-zombie-apocalypse>

Preparedness 101: Zombie Apocalypse

Posted February 01, 2013 by [Jamie-Leigh Cuthbertson](#) - Photo aficionado

With the release of a zombie rom-com movie this weekend and AMC's The Walking Dead returning to our television screens shortly, we thought it might be a great time to share our zombie preparedness tips. Fuelled by a blog about zombies and emergency preparedness from the [US Centers for Disease Control and Prevention](#), North America has become almost obsessed with imagining how they would fare if the walking dead started roaming their neighbourhood.



Photo Credit: <http://www.cdc.gov/phpr/zombies>

Did you know that your [emergency kit](#) contains tools and supplies that can also be used in the event of an outbreak of the zombie virus? Here is how items in your emergency kit can help you survive the zombie apocalypse:

Form of *help*

- The base form *help* occurs the most frequently with the bare infinitive.
- The form *helping* shows the highest proportion of *to*-infinitives.

*It is his job to see through the contracts that will **help rebuild** Iraq.*

*Many are partners with South African companies on projects that are **helping to rebuild** the country's infrastructure.*

Distance between *help* and Infinitive

- The more words between help and the infinitive, the higher the chances of the *to*-infinitive:

*HUD will provide \$70 million for 1,300 rental vouchers to **help** people in public housing projects in Los Angeles, Chicago, Boston, Baltimore, and New York **to move** into surrounding middle-class and affluent suburbs.*

- This tendency can be explained by the principle of minimization of cognitive complexity: v“[i]n the case of more or less explicit grammatical options the more explicit one(s) will tend to be favoured in cognitively more complex environments” (Rohdenburg 1996: 151).

Horror aequi

- A universal tendency to avoid repetition of identical elements.
- If the verb *help* is preceded by *to*, the second infinitive is usually without *to*:

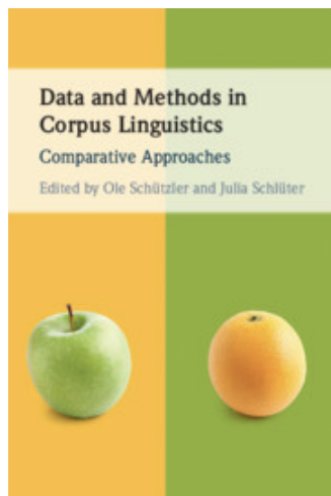
*With yoga, find a teacher who will make adjustments to **help prevent** injuries.*

Helpee

- The chances of the marked infinitive are higher if the Helpee is implicit:

*[...] physical therapy has **helped** me **improve** my posture...*

*His encouragement and guidance **helped** [Ø] **to improve** my health and self-confidence.*



Data and Methods in
Corpus Linguistics
Comparative
Approaches

8 - Comparing Bayesian and Frequentist Models of Language Variation

The Case of Help + (to-)Infinitive

from Part III - Perspectives on Multifactorial Methods

Published online by Cambridge University Press: **06 May 2022**

By [Natalia Levshina](#)

Edited by [Ole Schützler](#) and [Julia Schlüter](#)

[Show author details](#) ▾

Chapter

Data: "Big" dataset

- 2050 examples from COCA
- Random effects: Verb
- Fixed effects:
 - Year_new (transformed Year)
 - Horror: are there conditions for horror aequi effects?
 - Distance_log: distance in words between help and the infinitive, log-transformed
 - Morph_Form: morphological form of help
 - Helpee: the presence or absence of overt Helpee
 - Helper: whether the Helper is animate, inanimate or missing.

Read the dataset in R

```
help_df <- read.table("help_big.txt", header = T,  
stringsAsFactors = TRUE)
```

```
str(help_df)
```

```
'data.frame': 2050 obs. of 10 variables:  
 $ Year      : int  2011 2002 1998 2009 2009 2001 2006 2006 2006 2003 ...  
 $ Response  : Factor w/ 2 levels "Bare","TO": 1 2 1 1 1 1 1 2 1 1 ...  
 $ Horror    : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 2 2 1 1 1 ...  
 $ MorphForm : Factor w/ 4 levels "help","helped",...: 1 1 1 1 1 1 1 1 1 1 ...  
 $ Distance  : int   6 6 6 5 5 5 4 4 4 4 ...  
 $ Helpee    : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 2 ...  
 $ Verb      : Factor w/ 594 levels "abate","absorb",...: 421 223 315 283 309  
38 454 351 195 361 ...  
 $ Helper    : Factor w/ 3 levels "Anim","Inanim",...: 3 1 2 2 3 3 1 1 2 2 ...  
 $ Year_new   : int   21 12 8 19 19 11 16 16 16 13 ...  
 $ Distance_log: num   1.79 1.79 1.79 1.61 1.61 ...
```

Fitting a glmer model

```
library(lme4)
```

```
help_glmer <- glmer(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helppee + (1|Verb), data  
= help_df, family = binomial)
```

```
#convergence issues
```

```
help_glmer <- glmer(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helppee + (1|Verb), data  
= help_df, family = binomial,  
glmerControl(optimizer="bobyqa", optCtrl = list(maxfun =  
100000)))
```

summary(help_glmer)

...

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.08956	0.17217	-6.328	2.48e-10	***
Year_new	-0.05661	0.01046	-5.415	6.14e-08	***
HorrorYes	-6.46193	1.56818	-4.121	3.78e-05	***
Distance_log	0.85528	0.41933	2.040	0.04139	*
MorphFormhelped	0.50307	0.19266	2.611	0.00902	**
MorphFormhelping	3.14587	0.28112	11.190	< 2e-16	***
MorphFormhelps	1.25889	0.24905	5.055	4.31e-07	***
HelpeeYes	-0.99159	0.39751	-2.495	0.01261	*
HorrorYes:Distance_log	4.16950	1.33173	3.131	0.00174	**
MorphFormhelped:HelpeeYes	0.72105	0.37751	1.910	0.05613	.
MorphFormhelping:HelpeeYes	-1.80414	0.40270	-4.480	7.46e-06	***
MorphFormhelps:HelpeeYes	0.23383	0.42164	0.555	0.57918	

C-index: measure of discriminative ability

- Concordance index C
- Recommended for models with binary responses (e.g., logistic models, classification trees, random forests).
- If you take all possible pairs that contain a sentence with outcome A and a sentence with outcome B , the C-index will be the proportion of the times when the model predicts a higher probability of A for the sentence with A , and a higher probability of B for the sentence with B (Levshina, 2015: 259).

Interpretation of different values of C-index

- **0.5:** The model is no better than random chance (flipping a coin)
- **0.7 - 0.8:** acceptable
- **0.8 - 0.9:** good
- **0.9 - 1.0:** excellent

Compute C-index

- We need predicted probabilities of TO-infinitive, for every observation.

```
glmer_predict <- predict(help_glmer, type = "response")  
head(glmer_predict)
```

1	2	3	4	5	6
0.3110800	0.4093780	0.2579603	0.1551730	0.1947764	0.2440183

```
length(glmer_predict)
```

```
[1] 2050
```

```
library(Hmisc)
somers2(glmer_predict, as.numeric(help_df$Response) - 1)

      C      Dxy      n      Missing
0.8531138 0.7062275 2050.0000000 0.0000000
```

For this model, $C = 0.854$. This means that for 85.3% of all possible pairs of examples with *to*- and bare infinitive, the predicted probability of *to*-infinitive is higher for the sentence where the speaker actually used the *to*-infinitive than for the example where the bare infinitive occurred.

Bayesian GLMM

```
library(brms)
```

A useful shortcut:

```
my_formula <- bf(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helpee + (1|Verb))
```

The default priors

```
get_prior(formula = my_formula, data = help_df, family =
bernoulli)
```

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
(flat)	b									default
(flat)	b	Distance_log								(vectorized)
(flat)	b	HelpeeYes								(vectorized)
(flat)	b	HorrorYes								(vectorized)
(flat)	b	HorrorYes:Distance_log								(vectorized)
(flat)	b	MorphFormhelped								(vectorized)
(flat)	b	MorphFormhelped:HelpeeYes								(vectorized)
(flat)	b	MorphFormhelping								(vectorized)
(flat)	b	MorphFormhelping:HelpeeYes								(vectorized)
(flat)	b	MorphFormhelps								(vectorized)
(flat)	b	MorphFormhelps:HelpeeYes								(vectorized)
(flat)	b	Year_new								(vectorized)
student_t(3, 0, 2.5)	Intercept									default
student_t(3, 0, 2.5)	sd						0			default
student_t(3, 0, 2.5)	sd		Verb				0			(vectorized)
student_t(3, 0, 2.5)	sd	Intercept	Verb				0			(vectorized)

Fitting a GLMM with weakly informative priors

```
my_priors <- prior(cauchy(0, 2), class = b) +  
  prior(normal(0, 1), class = sd)
```

```
help_brm <- brm(formula = my_formula, data = help_df,  
prior = my_priors, family = bernoulli, cores = 4)
```

Compare the betas

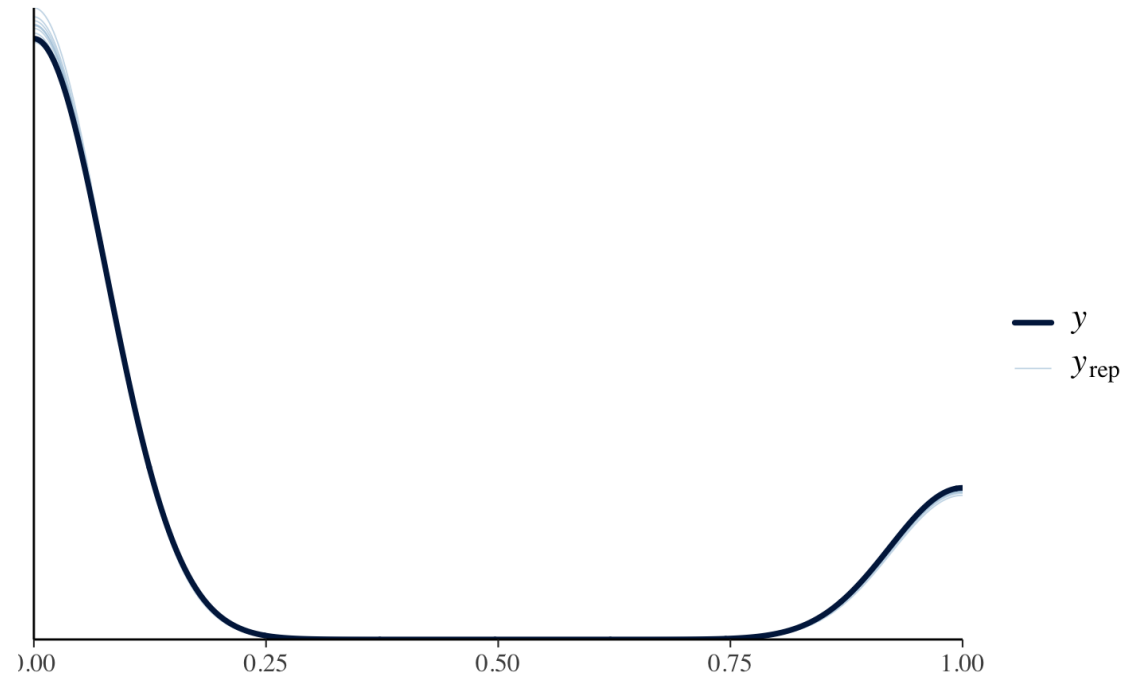
```
cbind(glmer = fixef(help_glmer), brm = fixef(help_brm)[, 1])
```

	glmer	brm
(Intercept)	-1.08955621	-1.09177319
Year_new	-0.05661283	-0.05661474
HorrorYes	-6.46192823	-5.60633604
Distance_log	0.85527870	0.83163250
MorphFormhelped	0.50306637	0.49880015
MorphFormhelping	3.14587162	3.11019739
MorphFormhelps	1.25888931	1.24317773
HelpeeYes	-0.99159035	-0.98245119
HorrorYes:Distance_log	4.16950112	3.18564454
MorphFormhelped:HelpeeYes	0.72105316	0.70749160
MorphFormhelping:HelpeeYes	-1.80413945	-1.74856346
MorphFormhelps:HelpeeYes	0.23383437	0.23947953

Posterior predictive check

- The traditional check for LM is probably not very meaningful for binary outcomes.

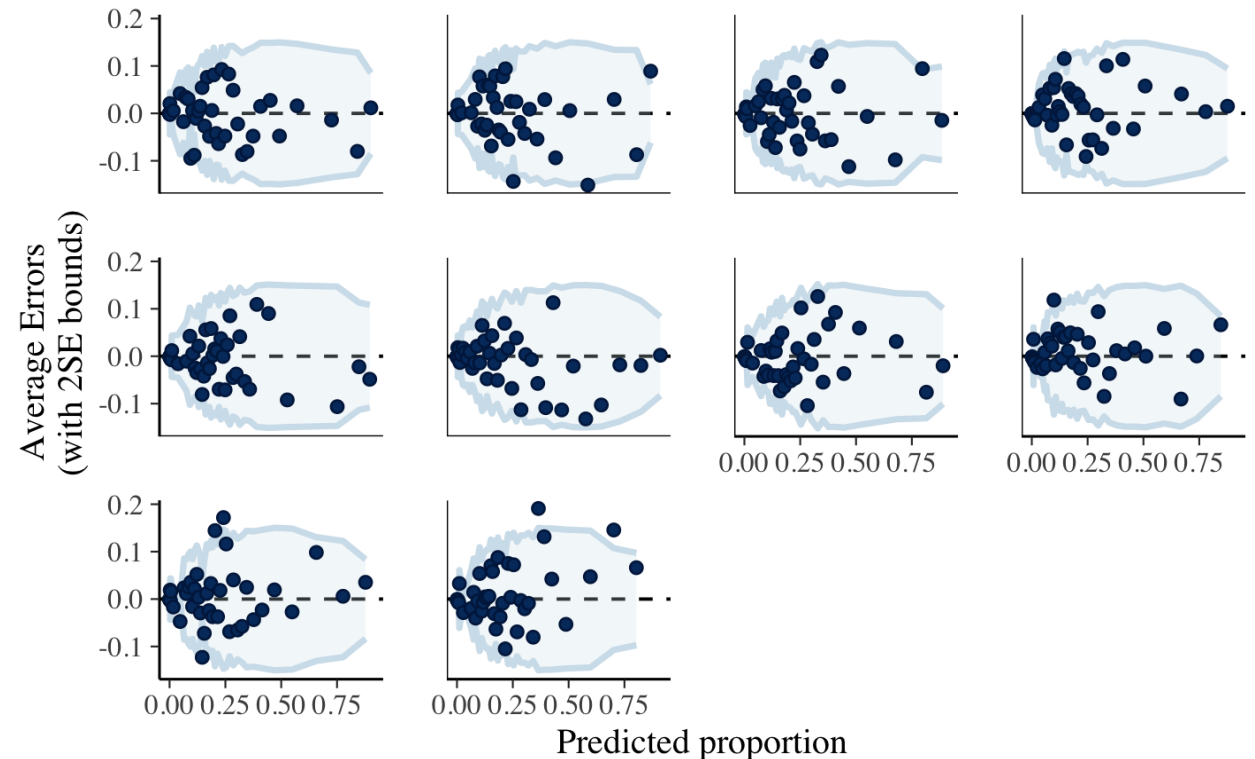
`pp_check(help_brm)`



Posterior predictive check of binned errors

```
pp_check(help_brm, type = "error_binned")
```

- Points: bins of data points. Data points that the model assigned similar predictions to will be binned together.
- X-axis: predicted proportions of TO
- Y-axis: errors
- 95% of points should be within the blue regions



Bayesian C-index

- To compute the C-index, we need the predicted probabilities of TO-infinite for every observation in the dataset (N = 2050)...
- AND for every posterior draw (N = 4000), because we are doing Bayesian statistics!

```
preds <- posterior_linpred(help_brm, newdata = help_df,  
transform = TRUE)
```

```
dim(preds)
```

```
[1] 4000 2050
```

Probabilities of TO-infinitive

A fragment of the matrix:

`preds[1:5, 1:5]`

	[, 1]	[, 2]	[, 3]	[, 4]	[, 5]
[1,]	0.8179976	0.8494583	0.44975237	0.08342518	0.5000209
[2,]	0.1778396	0.2292900	0.22343480	0.34528501	0.1421977
[3,]	0.2730770	0.4466109	0.54543907	0.12741657	0.1790101
[4,]	0.5849022	0.5252031	0.09837382	0.30683309	0.4291105
[5,]	0.1462877	0.1402333	0.16004696	0.14338837	0.1557020

Computing the Bayesian C-score

```
library(pROC)
for (i in 1:4000) {
  c_obj <- roc(response = help_df$Response, predictor =
preds[i, ], quiet = TRUE)
  c_scores[i] <- auc(c_obj)
}
mean(c_scores)
[1] 0.837815
quantile(c_scores, probs = c(0.025, 0.975))
2.5%      97.5%
0.8201381 0.8545324
```

A practical challenge

- To incorporate a reviewer's suggestion, I decided to annotate the data for a very time-consuming variable, Stress (clash, lapse or nothing).
- Unfortunately, this was very time-consuming and difficult. I only managed to go through 400 examples.
- The new dataset may be too small...

Horrors of horror aequi

```
help_glmer_small <- glmer(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helpee + Stress +  
(1|Verb), data = help_df_small, family = binomial,  
glmerControl(optimizer="bobyqa", optCtrl = list(maxfun =  
100000)))
```

```
summary(help_glmer_small)
```

...

HorrorYes:Distance_log	-21.65925	3460.01523	-0.006	0.995005
------------------------	-----------	------------	--------	----------

And other horrors...

```
optimizer (bobyqa) convergence code: 0 (OK)
```

```
unable to evaluate scaled gradient
```

```
Model failed to converge: degenerate Hessian with 2 negative  
eigenvalues
```

```
See ?lme4::convergence and ?lme4::troubleshooting.
```

```
Warning messages:
```

```
1: In vcov.merMod(object, use.hessian = use.hessian) :
```

```
variance-covariance matrix computed from finite-difference Hessian is  
not positive definite or contains NA values: falling back to var-cov  
estimated from RX
```

```
2: In vcov.merMod(object, correlation = correlation, sigm = sig) :
```

```
variance-covariance matrix computed from finite-difference Hessian is  
not positive definite or contains NA values: falling back to var-cov  
estimated from RX
```

Data separation

```
table(help_df_small$Horror, help_df_small$Distance,  
help_df_small$Response)
```

```
, , = Bare
```

	1	2	3	4	5
No	136	71	7	1	0
Yes	52	45	2	1	0

```
, , = TO
```

	1	2	3	4	5
No	64	13	4	1	2
Yes	1	0	0	0	0

Bayesian models with different priors

- Non-informative, default priors for betas:

```
help_brm_flat <- brm(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helped + Stress +  
(1|Verb), data = help_df_small, family = bernoulli,  
cores = 4)
```

- Weakly informative priors for betas (see above):

```
help_brm_weak <- brm(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helped + Stress +  
(1|Verb), data = help_df_small, family = bernoulli,  
prior = my_priors, cores = 4)
```

Bayesian model with strong priors

- The priors for the problematic interaction have the mean from the posterior means and standard deviation from Est. error

```
my_priors_strong <- prior(cauchy(0, 2), class = "b") +  
prior(normal(4.5, 1.5), class = "b", coef =  
"HorrorYes:Distance_log")
```

```
help_brm_strong <- brm(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helpee + Stress +  
(1|Verb), data = help_df_small, family = bernoulli,  
prior = my_priors_strong, cores = 4)
```

Compare the coefficients in six models

```
cbind(big_glmer = fixef(help_glmer),  
      big_brms = fixef(help_brms)[, 1],  
      small_glmer = fixef(help_glmer_small)[-c(9:10)],  
      small_brms_flat = fixef(help_brms_flat)[-c(9:10), 1],  
      small_brms_weak = fixef(help_brms_weak)[-c(9:10), 1],  
      small_brms_strong = fixef(help_brms_strong)[-c(9:10), 1])
```

What is the effect of the strong prior?

Where do you still see the greatest discrepancies?

Exercise

Try to add another strong prior based on the big dataset model. Will your choice of priors influence the posteriors?